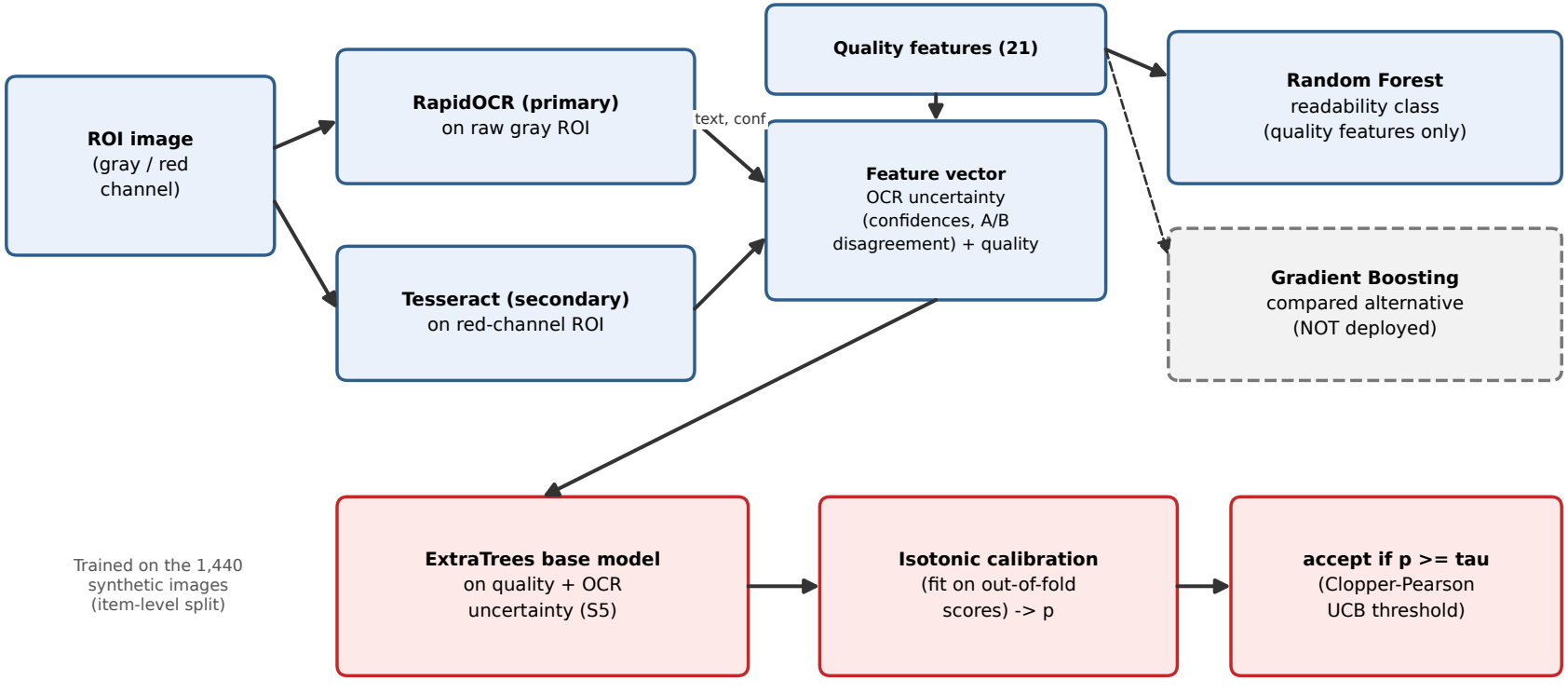
OCR and machine-learning workflow: which model does what
ROI image
(gray / red
channel)
RapidOCR (primary)
on raw gray ROI
Tesseract (secondary)
on red-channel ROI
text, conf
Quality features (21)
Feature vector
OCR uncertainty
(confidences, A/B
disagreement) + quality
Random Forest
readability class
(quality features only)
Gradient Boosting
compared alternative
(NOT deployed)
Trained on the 1,440
synthetic images
(item-level split)
ExtraTrees base model
on quality + OCR
uncertainty (S5)
Isotonic calibration
(fit on out-of-fold
scores) -> p
accept if p >= tau
(Clopper-Pearson
UCB threshold)